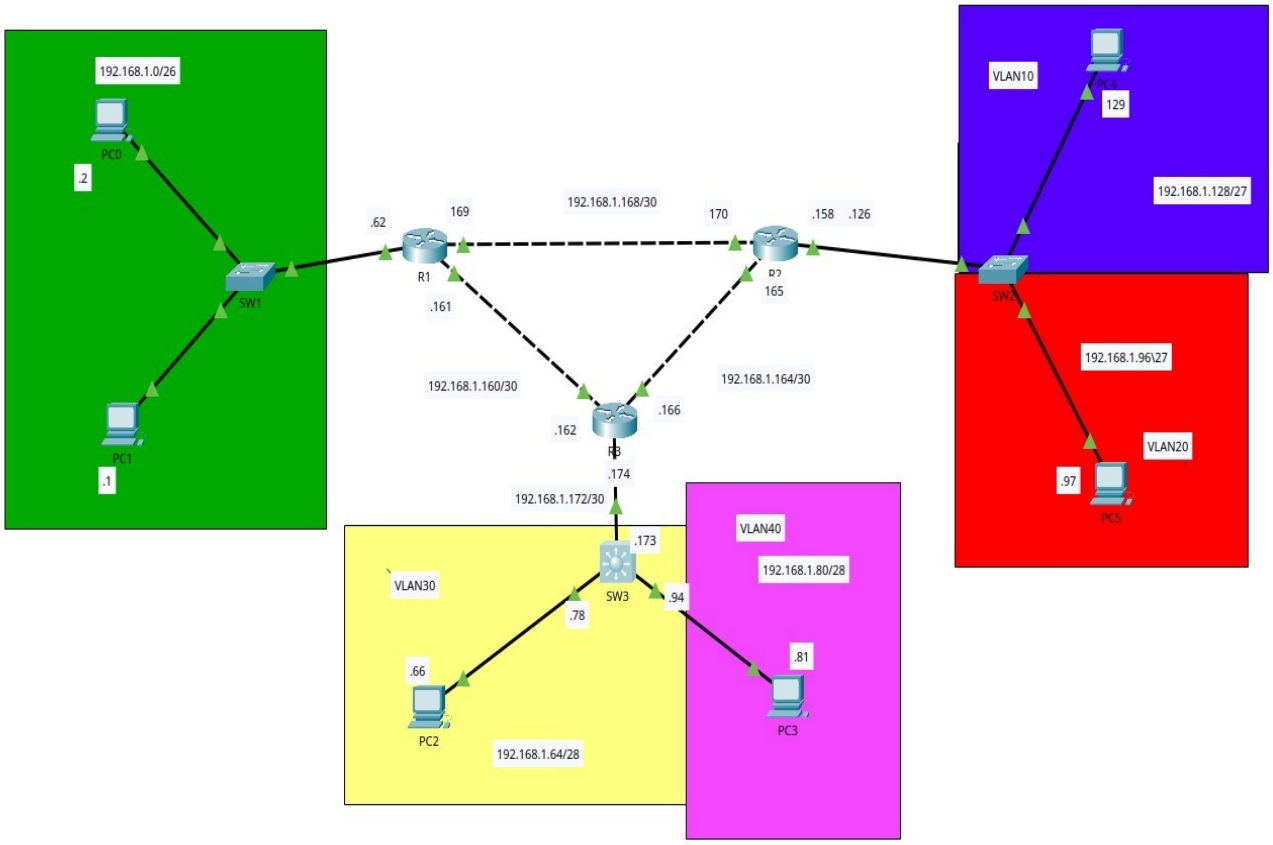
Scheme



The network topology is basic. It features 3 routers. After all of them is a Switch. The left switch has no configurations. The other 2 switches use VLANs. The right switch uses simple trunk port to the router. The switch in the middle is multilayer switch.

Router Configuration

This is the configuration for the right router. It features the sub-interface method to connect to the VLANs via ROAS. The other ones follow the same pattern, except they don't have sub-interfaces configured, only physical.

```
Router>en\
Translating "en\"...domain server (255.255.255.255) % Name lookup aborted
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R2
R2(config)#int g0/0
R2(config-if)#int g0/0.10
R2(config-subif)#enc
R2(config-subif)#encapsulation dot
R2(config-subif)#encapsulation dot1Q 10
R2(config-subif)#ip add 192.168.1.158 255.255.255.224
R2(config-subif)#no sh
R2(config-subif)#int g0/0.20
R2(config-subif)#ip add 192.168.1.126 255.255.255.224

% Configuring IP routing on a LAN subinterface is only allowed if that
subinterface is already configured as part of an IEEE 802.10, IEEE 802.1Q,
or ISL vLAN.

R2(config-subif)#encapsulation d
R2(config-subif)#encapsulation dot1Q 20
R2(config-subif)#ip add 192.168.1.126 255.255.255.224
R2(config-subif)#no sh
R2(config-subif)#
R2(config-subif)#int g0/2
R2(config-if)#ip add 192.168.1.165 255.255.255.252
R2(config-if)#no sh

R2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

R2(config-if)#int g0/1
R2(config-if)#ip add 192.168.1.170 255.255.255.252
R2(config-if)#no sh

R2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
```

Trunk port configuration on right switch

```
1    default                          active    Fa0/3, Fa0/4, Fa0/5, Fa0/6
                                               Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                               Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                               Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                               Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                               Fa0/23, Fa0/24, Gig0/2
10   VLAN0010                         active    Fa0/1
20   VLAN0020                         active    Fa0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

```
Enter configuration commands, one per line.  End with CNTL/Z.
SW2(config)#int g0/0
%Invalid interface type and number
SW2(config)#swi
SW2(config)#swit
SW2(config)#int g0/0
%Invalid interface type and number
SW2(config)#int g0/1
SW2(config-if)#swit
SW2(config-if)#switchport mo
SW2(config-if)#switchport mode tr
SW2(config-if)#switchport mode trunk
SW2(config-if)#switchport tru
SW2(config-if)#switchport trunk al
SW2(config-if)#switchport trunk allowed vl
SW2(config-if)#switchport trunk allowed vlan 10,20
SW2(config-if)#do sh int trunk
```

This router uses simple trunk. It has the 2 VLANs connected directly to it through access ports.

Middle switch

Since this is a multilayer switch it's configuration is like the one for a router.

```
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname SW3
SW3(config)#ip routing
SW3(config)#int g0/0
%Invalid interface type and number
SW3(config)#int g0/1
SW3(config-if)#no switchport
SW3(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

SW3(config-if)#ip add 192.168.1.173 255.255.255.252
SW3(config-if)#no sh
```

For the connection between this switch and the router I used default route on the switch to the router, and static routes on the router

```
SW3#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 192.168.1.174 to network 0.0.0.0

     192.168.1.0/24 is variably subnetted, 3 subnets, 2 masks
C       192.168.1.64/28 is directly connected, FastEthernet0/1
C       192.168.1.80/28 is directly connected, FastEthernet0/2
C       192.168.1.172/30 is directly connected, GigabitEthernet0/1
S*   0.0.0.0/0 [1/0] via 192.168.1.174

SW3#
```

```
R3#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     192.168.1.0/24 is variably subnetted, 12 subnets, 5 masks
O       192.168.1.0/26 [110/2] via 192.168.1.161, 00:10:43, GigabitEthernet0/1
S       192.168.1.64/28 [1/0] via 192.168.1.173
S       192.168.1.80/28 [1/0] via 192.168.1.173
```

Routing - RIP

I did the routing between LANs with RIP and OSPF. This is a routing table using RIP.

```
R2>en
R2#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     192.168.1.0/24 is variably subnetted, 13 subnets, 5 masks
R       192.168.1.0/26 [120/1] via 192.168.1.169, 00:00:04, GigabitEthernet0/1
R       192.168.1.64/28 [120/2] via 192.168.1.166, 00:00:06, GigabitEthernet0/2
R       192.168.1.80/28 [120/2] via 192.168.1.166, 00:00:06, GigabitEthernet0/2
C       192.168.1.96/27 is directly connected, GigabitEthernet0/0.20
L       192.168.1.126/32 is directly connected, GigabitEthernet0/0.20
C       192.168.1.128/27 is directly connected, GigabitEthernet0/0.10
L       192.168.1.158/32 is directly connected, GigabitEthernet0/0.10
R       192.168.1.160/30 [120/1] via 192.168.1.166, 00:00:06, GigabitEthernet0/2
                         [120/1] via 192.168.1.169, 00:00:04, GigabitEthernet0/1
C       192.168.1.164/30 is directly connected, GigabitEthernet0/2
L       192.168.1.165/32 is directly connected, GigabitEthernet0/2
C       192.168.1.168/30 is directly connected, GigabitEthernet0/1
L       192.168.1.170/32 is directly connected, GigabitEthernet0/1
R       192.168.1.172/30 [120/1] via 192.168.1.166, 00:00:06, GigabitEthernet0/2

R2#
```

Routing - OSPF

After configuring with RIP I removed it and configured OSPF on the same topology. It has only 1 area. This is the routing table of R3 with OSPF.

```
     192.168.1.0/24 is variably subnetted, 12 subnets, 5 masks
O       192.168.1.0/26 [110/2] via 192.168.1.161, 00:38:05, GigabitEthernet0/1
S       192.168.1.64/28 [1/0] via 192.168.1.173
S       192.168.1.80/28 [1/0] via 192.168.1.173
O       192.168.1.96/27 [110/2] via 192.168.1.165, 00:35:55, GigabitEthernet0/2
O       192.168.1.128/27 [110/2] via 192.168.1.165, 00:36:05, GigabitEthernet0/2
C       192.168.1.160/30 is directly connected, GigabitEthernet0/1
L       192.168.1.162/32 is directly connected, GigabitEthernet0/1
C       192.168.1.164/30 is directly connected, GigabitEthernet0/2
L       192.168.1.166/32 is directly connected, GigabitEthernet0/2
O       192.168.1.168/30 [110/2] via 192.168.1.161, 00:37:00, GigabitEthernet0/1
                         [110/2] via 192.168.1.165, 00:37:00, GigabitEthernet0/2
C       192.168.1.172/30 is directly connected, GigabitEthernet0/0
L       192.168.1.174/32 is directly connected, GigabitEthernet0/0
```

Also configured the router IDs. This are the neighbours of R3.

```
R3#sh ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.1           1   FULL/BDR        00:00:34    192.168.1.161   GigabitEthernet0/1
2.2.2.2           1   FULL/BDR        00:00:31    192.168.1.165   GigabitEthernet0/2
```

And since the R3 has static routes to networks that are not directly connected I had to use the command ‘redistribute static subnets’ to get the router to advertise them.